

COBRA Machine Learning

User Manual

*Comprehensive Distance Based Ranking integrated with
Machine Learning, Gaussian Balancing and PSI*

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INPI Registration No. BR512026001134-5
Version 2.0 — 2026*

1. Introduction

COBRA Machine Learning (COBRA-ML) is a decision support tool that combines the COBRA (Comprehensive Distance Based Ranking) multi-criteria method with Machine Learning techniques, Gaussian weight balancing, and the PSI (Preference Selection Index) method.

The tool is designed for researchers, analysts, and decision-makers who need to rank alternatives based on multiple criteria, especially when:

- Criteria weights are not pre-defined or objective weight calculation is desired
- Automated parameter tuning through machine learning is required
- The problem involves trade-offs between benefit criteria (maximize) and cost criteria (minimize)

COBRA-ML is a Streamlit-based web application, bilingual (Portuguese/English), running locally or on any server that supports Python.

1.1 Key Features

- Multi-criteria ranking using the COBRA method with Euclidean distance
- Objective weight calculation via PSI (Preference Selection Index)
- Gaussian balancing of weights with parameters λ (λ), μ (μ), and σ (σ)
- Automatic parameter prediction using a Random Forest Regressor

- Three built-in sample problems: Renewable Energy, Fighter Aircraft, and Suppliers (read-only)
- Support for custom problems (fully editable) and file upload (Excel/CSV)
- Synthetic dataset generator for ML training and empirical validation
- Interactive visualizations: heatmaps, radar charts, Gaussian function plots
- Export to CSV, Excel, and JSON
- Optional AI-powered analysis via compatible AI API (OpenAI-compatible)
- Bilingual interface (Portuguese and English)

1.2 Application Areas

- Renewable energy source selection
- Military equipment acquisition and evaluation
- Supplier selection and procurement
- Technology assessment
- Any multi-criteria decision-making (MCDM) problem

2. Installation and Setup

2.1 System Requirements

Requirement	Version / Details
Python	3.8 or higher
Streamlit	1.x
scikit-learn	1.x
pandas	1.x or 2.x
numpy	1.x
plotly	5.x
openai (optional)	1.x (for AI features)
xlsxwriter	latest (for Excel export)

2.2 Installing Dependencies

Open a terminal and run:

```
pip install streamlit pandas numpy scikit-learn plotly openpyxl xlsxwriter
```

To enable the optional AI analysis feature:

```
pip install openai python-dotenv
```

2.3 Running the Application

1. Save the file `cobra_ml_app_integrated.py` to your working directory.
2. Open a terminal in that directory.
3. Run the following command:

```
streamlit run cobra_ml_app_integrated.py
```

- The application will open automatically at <http://localhost:8501>

Tip: If the browser does not open automatically, navigate manually to <http://localhost:8501> in your browser.

3. Application Interface

3.1 Language Selection

At the top of the application there is a language selector. You can switch between Portuguese and English at any time. The interface, labels, and output descriptions update immediately.

3.2 Sidebar Settings

Setting	Description
Data Source	Choose from built-in examples (read-only) or provide your own data
Use Machine Learning	Enables ML-based prediction of optimal parameters (λ , μ , σ)
Train ML Model	Generates synthetic data and trains a new Random Forest model
Number of Training Samples	Controls how many synthetic problems are used for training (10–100)
AI API Key	Optional — enables AI-powered result analysis (OpenAI-compatible key)
Clear Results	Resets all computed results and returns to initial state

3.3 Data Sources

Mode	Description	Editable?	When to Use
Renewable Energy	5 alternatives, 5 criteria (built-in)	Read-only	Quick demo or energy studies
Fighter Aircraft	5 alternatives, 6 criteria (built-in)	Read-only	Military or procurement cases
Suppliers	4 alternatives, 5 criteria (built-in)	Read-only	Supply chain analysis
Custom Problem	User-defined alternatives and criteria	Fully editable	Any new decision problem
Upload File	Excel (.xlsx) or CSV file	Criteria types editable	Pre-existing datasets

Note: Built-in examples (Renewable Energy, Fighter Aircraft, Suppliers) are read-only and serve as demonstrations only. To analyse your own problem, select Custom Problem or Upload File.

4. Step-by-Step Usage Guide

4.1 Using a Built-in Example

5. Select a Data Source in the sidebar (e.g., Renewable Energy).
6. The decision matrix and criteria types are displayed as read-only in the main panel.
7. Criteria types (Benefit/Cost) are shown as informational badges — they cannot be changed for built-in examples.
8. Optionally enable Machine Learning in the sidebar.
9. Click the Run COBRA-ML Analysis button.
10. View the results in the tabs: Final Ranking, Weights Analysis, Matrices, ML Parameters, and Visualizations.

Built-in examples use fixed data and criteria types that match the published article. If you need different values, use Custom Problem or Upload File.

4.2 Using a Custom Problem

11. Select Custom Problem in the Data Source dropdown.
12. Set the Number of Alternatives and Number of Criteria.
13. Enter the names of your alternatives in the text fields.
14. Enter criterion names and select their type (Benefit or Cost) for each.
15. Fill in the decision matrix values.
16. Review and adjust criteria types in the Criteria Information expander if needed.
17. Click Run COBRA-ML Analysis.

Benefit criteria are those you want to maximize (e.g., quality, efficiency). Cost criteria are those you want to minimize (e.g., price, emissions).

4.3 Uploading a File

Your file must follow this format:

- First column: alternative names (labeled Alternatives or similar)
- Remaining columns: criteria values, one column per criterion
- First row: column headers

Example Excel structure:

Alternatives	Cost (\$)	Quality (1-10)	Delivery (days)
Option A	1000	8	15
Option B	1200	9	10
Option C	900	7	20

After uploading, configure the criteria types (Benefit/Cost) in the Criteria Information expander before running the analysis.

4.4 Configuring Machine Learning Options

Mode	Description	When to Use
ML with trained model	Uses existing model to predict λ , μ , σ from problem features	When model has been previously trained

ML + Train new model	Generates synthetic training data and retrains the Random Forest	First use or when updating the model
Manual parameters	User defines λ , μ , and σ via sliders in the sidebar	When domain expertise is available

5. Methodology

5.1 Overview of COBRA-ML

Component	Role
COBRA	Core ranking using distances to Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS)
PSI (Preference Selection Index)	Objective weight calculation based on variance of normalized criteria values
Gaussian Balancing	Adjusts weights using a Gaussian bell-curve parametrized by λ , μ , and σ
Random Forest Regressor	Predicts optimal λ , μ , and σ from problem structure features automatically

5.2 Method Execution Flow

When you click Run COBRA-ML Analysis, these steps are executed:

18. Decision Matrix Normalization: each criterion column is normalized using vector normalization. Benefit criteria use the column vector; cost criteria use the inverse values before normalization.
19. PSI Weights Calculation: the Preference Selection Index computes objective weights from the variance of normalized values. Criteria with lower relative dispersion receive higher weight.
20. Parameter Prediction or Manual Input: if ML is enabled, the Random Forest model predicts λ , μ , σ from features extracted from the decision matrix (m, n, PSI weight variance, entropy, data dispersion, etc.).
21. Gaussian Weight Balancing: final weights are computed by combining PSI weights with a Gaussian distribution function, controlled by λ (blending), μ (center), and σ (spread).
22. COBRA Score Calculation: the weighted matrix is computed and for each alternative distances to PIS and NIS are calculated. Final COBRA score = $D^- / (D^+ + D^-)$.
23. Ranking: alternatives are ordered in descending order of COBRA-ML scores. The highest score is the best choice.

5.3 Gaussian Parameters Explained

Parameter	Symbol	Range	Effect
Lambda	λ	0.0 to 1.0	Blending: 0 = full PSI weights, 1 = full Gaussian-adjusted PSI
Mu	μ	0 to n-1	Center of the Gaussian bell curve — index of most emphasised criterion
Sigma	σ	0.5 to n	Spread of the bell curve — how many criteria are emphasised

6. Reading the Results

6.1 Final Ranking Tab

Displays a ranked table of all alternatives with their COBRA-ML scores. The alternative in Position 1 is the best according to the model and the given criteria. A bar chart visualises the scores.

Higher COBRA-ML scores indicate better alternatives. Scores range from 0 to 1.

6.2 Weights Analysis Tab

Shows two sets of weights for each criterion:

- PSI Weights (Objective): derived purely from data variability, without subjective input.
- Final Weights (Balanced): the result after Gaussian balancing — these are the actual weights used in scoring.

A grouped bar chart compares PSI vs final weights across criteria.

6.3 Matrices Tab

- Normalized Matrix (R): the decision matrix after vector normalization.
- Weighted Matrix (V): the normalized matrix multiplied by the final weights.

6.4 ML Parameters Tab

Parameter	Description
lambda (λ)	Weight blending factor between PSI and Gaussian weighting
mu (μ)	Center of the Gaussian emphasis on criteria
sigma (σ)	Spread (bandwidth) of the Gaussian emphasis

If ML was used, a note confirms the parameters were predicted by the model. Otherwise, default or manually set values are shown. The tab also plots the Gaussian balancing function over the criteria.

6.5 Visualizations Tab

- Gaussian Balancing Function Plot: shows the bell-curve applied to criteria weights.
- Heatmap: colour-coded normalized matrix highlighting strong and weak performers.
- Radar Chart: normalised performance profile of a selected alternative across all criteria.

7. Synthetic Dataset for ML Training

After running an analysis, the Generate Synthetic Dataset for ML Training section is available as an expandable panel.

7.1 Purpose

The Machine Learning component of COBRA-ML uses a Random Forest Regressor to automatically predict the optimal parameters (λ , μ , σ) from the structural features of each MCDM problem. Empirically training and validating this model requires a dataset of MCDM problems with known optimal parameters — which the dataset generator provides.

7.2 How the Generator Works

The generator creates synthetic MCDM problems varying:

- Number of alternatives (m: 3–11) and criteria (n: 3–9)
- Statistical distribution of data (uniform, normal, log-normal, mixed)
- Benefit/cost criteria ratio (30–80% benefit)

For each problem, eight structural features (f_1 – f_8) are extracted and optimal parameters are derived from heuristic rules validated in the literature:

- λ : based on the normalised Shannon entropy of PSI weights — more uniform weights → higher objectivity
- μ : centred on the criterion with the highest PSI weight
- σ : proportional to the number of criteria and weight uniformity

7.3 Dataset Columns

Column	Description
problem_id	Unique identifier for each synthetic problem
m	Number of alternatives
n	Number of criteria
benefit_ratio	Proportion of benefit criteria (f_8)
psi_weight_variance	Variance of PSI weights (f_3)
psi_weight_entropy	Shannon entropy of PSI weights (f_5)
data_std	Global standard deviation of data (f_6)
data_mean	Global mean of data (f_7)
mean_adj_variation	Mean variation between adjacent alternatives (f_4)
lambda_opt	Optimal λ parameter (ML target)
mu_opt	Optimal μ parameter (ML target)
sigma_opt	Optimal σ parameter (ML target)

7.4 Using the Dataset

24. Set the number of problems (20–500) and a random seed for reproducibility.
25. Click Generate Dataset and download the CSV.
26. In Python, use feature columns (m, n, benefit_ratio, psi_weight_variance, psi_weight_entropy, data_std, data_mean, mean_adj_variation) as X.
27. Use [lambda_opt, mu_opt, sigma_opt] as y (multi-output regression).
28. Train a RandomForestRegressor with an 80/20 hold-out split and evaluate with R^2 and MAE.

Recommended minimum: 100 problems for training. Use the seed field for reproducible experiments across different runs.

8. Exporting Results

After running an analysis the Export Results section becomes available. Three formats are supported:

Format	Content	Use Case
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CSV	Ranking, scores, PSI weights, final weights	Quick tabular analysis in Excel or Python
Excel (.xlsx)	Ranking sheet + Normalized Matrix + Weighted Matrix	Full structured report
JSON	Ranking, scores, model parameters, method label	Integration with other systems

Click the corresponding Download button to save the file locally.

9. AI-Powered Analysis (Optional)

COBRA-ML includes an optional integration with an AI API (OpenAI-compatible, using the GPT-4o-mini model) for automated interpretation of results.

9.1 Configuration

29. Obtain an API key from your AI provider (e.g., <https://platform.openai.com>).
30. Enter the key in the AI API Key field in the sidebar.
31. Alternatively, create a file named `code.env` in the application directory with:

```
OPENAI_API_KEY=your_key_here
```

9.2 Using AI Analysis

32. After running a COBRA-ML analysis, scroll to the AI Analysis section.
33. Click Generate AI Analysis.
34. The AI provides: result interpretation, weights analysis, evaluation of the winning alternative, and recommendations.
35. Use the Interactive AI Chat to ask follow-up questions about the results.

AI analysis is optional and requires an active API key. The AI receives only ranking scores, weights, and parameters — no personal or sensitive data is transmitted. Any OpenAI-compatible API key can be used.

10. Input File Format Reference

10.1 CSV Format

The first row must contain column headers. The first column should be alternative names:

```
Alternative, Criterion1, Criterion2, Criterion3 Alt_A, 45, 8, 120 Alt_B, 38, 7, 95 Alt_C, 52, 9, 110
```

10.2 Excel Format

Same structure as CSV. All data must be numeric (except the Alternatives column). Empty cells and non-numeric values may cause errors.

10.3 Common Errors and Solutions

Error	Cause	Solution
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Non-numeric values in matrix	Text or empty cells in data columns	Ensure all criterion values are numbers	
File not loading	Unsupported format	Use .csv or .xlsx only	
Wrong criteria types	All criteria default to Benefit	Review and change types in Criteria Information expander after uploading	
ML model not available	Model not yet trained	Enable Train ML Model in sidebar settings	
AI API error	Invalid or missing key	Check that the key is correct and has available credits	

11. How to Cite

If you use COBRA-ML in your research, please cite as follows:

11.1 APA Format

Portella, A. G. (Holder), Santos, M. dos, & Gomes, C. F. S. (Authors). (2026). COBRA Machine Learning [Computer software]. Registration No. BR512026001134-5. Instituto Nacional da Propriedade Industrial (INPI). Retrieved from <https://cobramachinelearning.streamlit.app/>

11.2 ABNT Format

PORTELLA, Anderson Gonçalves (Titular); SANTOS, Marcos dos; GOMES, Carlos Francisco Simões (Autores). COBRA Machine Learning. Registro de Programa de Computador nº BR512026001134-5. Rio de Janeiro: Instituto Nacional da Propriedade Industrial (INPI), 20 fev. 2026. Disponível em: <https://cobramachinelearning.streamlit.app/>.

12. Developers and Contact

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COBRA-ML was developed in the context of research in Production Engineering and Operations Research at Universidade Federal Fluminense (UFF) and Escola Naval, Brazil.

Appendix: Built-in Sample Problems

These problems are read-only in the application and serve as demonstrations. Values match the published article.

A. Renewable Energy Selection

Alternative	LCOE (\$/MWh)	Cap. Factor (%)	CO2 Emiss. (g/kWh)	Area (km ² /MW)	Tech Mat. (1-10)
Solar PV	45	25	50	4.0	9
Onshore Wind	38	35	12	0.5	9
Offshore Wind	65	45	10	0.3	7
Hydroelectric	55	50	24	15.0	10
Biomass	60	70	230	2.0	8

Criteria types: LCOE = Cost | Capacity Factor = Benefit | CO2 = Cost | Area = Cost | Tech Maturity = Benefit

B. Fighter Aircraft Selection

Alternative	Cost (M\$)	Speed (Mach)	Range (km)	Payload (kg)	Stealth	Maint.
F-35A	80	1.6	2200	8000	10	6
F-15EX	90	2.5	3900	13000	3	8
Rafale	85	1.8	3700	9500	5	7
Eurofighter	95	2.0	2900	7500	4	6
Su-57	100	2.0	3500	10000	8	4

Criteria types: Unit Cost = Cost | all others = Benefit

C. Supplier Selection

Alternative	Price (\$)	Quality	Delivery (d)	Sustainability	Reliability
Supplier A	100	8	15	7	9
Supplier B	120	9	10	9	8
Supplier C	90	7	20	6	7
Supplier D	110	8.5	12	8	8.5

Criteria types: Price = Cost | Delivery = Cost | Quality, Sustainability, Reliability = Benefit